

AMUTHEEZAN SIVAGNANAM

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U.S. Permanent Resident | [Scholar](#) | <https://www.linkedin.com/in/amutheezansivagnanam> | [Github](#) | Open to relocation

SUMMARY

- **PhD Candidate and Machine Learning Researcher** specializing in **Deep Reinforcement Learning**
 - **Six** years of academic research experience building **scalable reinforcement learning** and **deep learning** systems for efficient **ML training** and **deployment**
 - **Two** years of industrial experience in **software development**, **testing** and **CI/CD** while following **agile practices**
 - Good **publication** record with an **h-index of 6** and **100+ citations**, featuring in top-tier **AI and ML** conferences
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EDUCATION

Ph.D. Computer Science, (August 2025): **Pennsylvania State University**, University Park, PA, Advisor: Dr. Aron Laszka

- **Dissertation**: Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains

M.S. Computer Science, (August 2022): **University of Houston**, Houston, TX, Advisor: Dr. Aron Laszka

B.S. Computer Science and Engineering, (January 2018): **University of Moratuwa**, Moratuwa, Sri Lanka

RESEARCH EXPERIENCE

Graduate Research Assistant, *Pennsylvania State University*, (August 2022 - May 2025)

- Devised and evaluated a **Transformer** based **multi-agent reinforcement learning algorithm** to solve the **emergency responder reallocation problem** that **reduces** decision making time by **3 orders of magnitude**
- Leveraged a **deep reinforcement learning** approach to tackle the challenge of online vehicle routing with advance booking, enabling **prompt confirmation within seconds**

Graduate Research Assistant, *University of Houston*, (September 2019 - August 2022)

- Introduced a novel **reinforcement learning** based solution approach to solve the problem of **online booking for offline vehicle routing problem** which resulted in **20 % reduction of operational costs** for the transit agency
- Introduced **novel problem formulation and algorithms** that **lower the energy costs by \$140k** annually for public transit agencies operating mixed fleets of buses
- **Skills**: **Python**, **C++**, **Java**, **C**, **SQL**, **MySQL**, **MongoDB**, **SQLite**, **PyTorch**, **TensorFlow**, **Scikit-learn**, **Transformers**, **Machine Learning**, **Deep Learning**, **Computer Vision**, **Reinforcement Learning**, **DQN**, **DDPG**, **TRPO**, **PPO**, **GRPO**, **LLM**, **RLHF**, **RAG**, **Prompt Engineering**, **LangChain**, **Bayesian Optimization**, **Data Science**, **Data Visualization**, **Clustering**, **Statistical Analysis**, **Data Mining**, **SLURM**, **Ray**, **RLlib**, **OpenAI Gym**, **NumPy**, **pandas**, **matplotlib**, **Linear Programming**, **Integer Programming**, **CPLEX**, **Gurobi**, **Mosek**, **OR-Tools**, **Spark**, **Amazon Web Services (Sage Maker, S3, Lambda)**, **Git**, **Docker**, **Linux**, **Bash**, **Object-Oriented Development**, **MATLAB**

SELECT PUBLICATIONS

A. Sivagnanam et al.; Multi-Agent Reinforcement Learning with Hierarchical Coordination for Emergency Responder Stationing. International Conference on Machine Learning. **ICML 2024**. **Core Ranking A***

A. Sivagnanam et al.; Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit. International Joint Conference on Artificial Intelligence. **IJCAI 2022**. **Core Ranking A***, Invited Talk: Vienna Austria

A. Sivagnanam et al.; Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service. AAAI Conference on Artificial Intelligence. **AAAI 2021**. **Core Ranking A***, Invited Talk: Virtual Conference

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer, *LSEG Technology*, (January 2018 - July 2019)

- Introduced **unit testing** for libraries in the Post-Trade **C++** codebase, improving system performance and reliability
- Implemented and validated **Oracle database changes** for Post-Trade products for the Singapore Stock Exchange using **Behavior Driven Development (BDD)** testing approaches in **Java**, ensuring data integrity and system functionality
- Contributed to the **CI/CD pipeline** of the Post-Trade product using **Python** and **Git** ensuring seamless software updates

Software Engineering Intern, *WSO2 Lanka PVT Ltd*, (July 2016 - December 2016)

- Engineered an **alert generation system** to monitor disease outbreaks by analyzing descriptive **HL7/FHIR data**, automating email and SMS notifications, and supporting rapid decision-making through prompt data processing
- Developed a mechanism to evaluate hospital functionality by assessing bed and oxygen cylinder availability from admission and discharge messages, with **effective requirements gathering** and integration of **structured data processing**
- **Skills**: **Python**, **C++**, **JAVA**, **JS**, **PHP**, **SQL**, **Oracle DB**, **Object Oriented Development**, **Github**, **Linux**, **Agile Practices**, **Behaviour Driven Development**, **Selenium**, **Vaadin**, **Natural Language Processing**, **Data Science**, **Data Visualization**, **HL7**, **FHIR**, **CI/CD**

REFERENCES

Name	Dr. Aron Laszka (PhD Advisor)	Dr. Abhishek Dubey	Dr. Weidong Shi
Position	Assistant Professor	Associate Professor	Associate Professor
Affiliation	Pennsylvania State University	Vanderbilt University	University of Houston
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